

Numerical modeling of projectile penetration into compressible rigid visco-plastic media

Ioan R. IONESCU

(e-mails: ioan.ionescu@lpmtm.univ-paris13.fr)

LPMTM, University Paris 13

99, Avenue Jean-Baptiste Clement

F-93430 Villetaneuse, France

Oana CAZACU

(e-mails: cazacu@reef.eng.ufl.edu)

REEF, University of Florida

1350 N. Poquito Road

Shalimar FL 32579, USA

Thomas PERROT

(e-mails: thomas.perrot@wanadoo.fr)

LAMA, University of Savoie

Campus Scientifique

F-73376 Le Bourget-du-Lac, France

Abstract:

We develop here computational methods for modeling the penetration of a rigid projectile into porous media. Compressible rigid visco-plastic models (see [1]) are used to capture the solid-fluid transition in behavior at high strain rates, account for damage/plasticity couplings and viscous effects observed in geological and cementitious materials.

A hybrid time-discretization is used to model the non-stationary flow of the target material and the projectile-target interaction, i.e. an explicit Euler method for the projectile equation and a forward (implicit) method for the target boundary value problem. At each time step, a mixed finite-element and finite-volume strategy is used to solve the "target" boundary value problem. Specifically, the nonlinear variational inequality for the velocity field is discretized using the finite element method while a finite volume method is adopted for the hyperbolic mass conservation and damage evolution equations. To solve the velocity problem, a decomposition-coordination formulation coupled with the augmented lagrangian method is adopted.

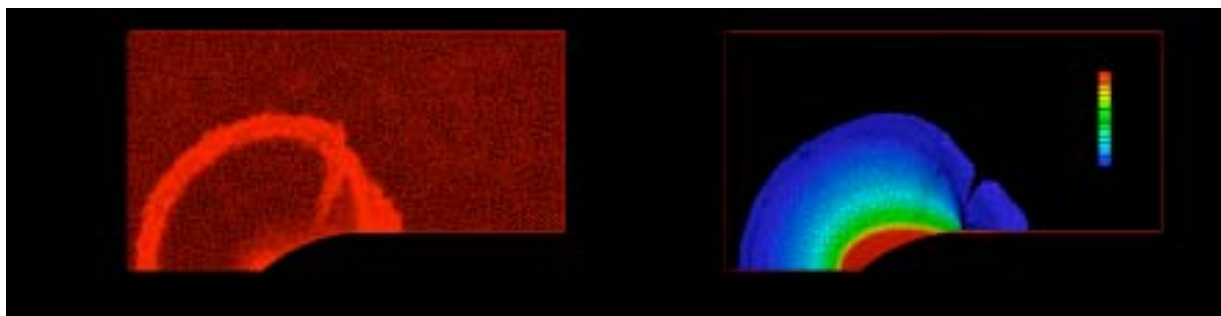


Figure 1. The finite element mesh obtained after remeshing (left), and the distribution of the deformation rate in 1/sec (right). Note that the boundary between the rigid zone and the viscoplastic one is sharply captured.

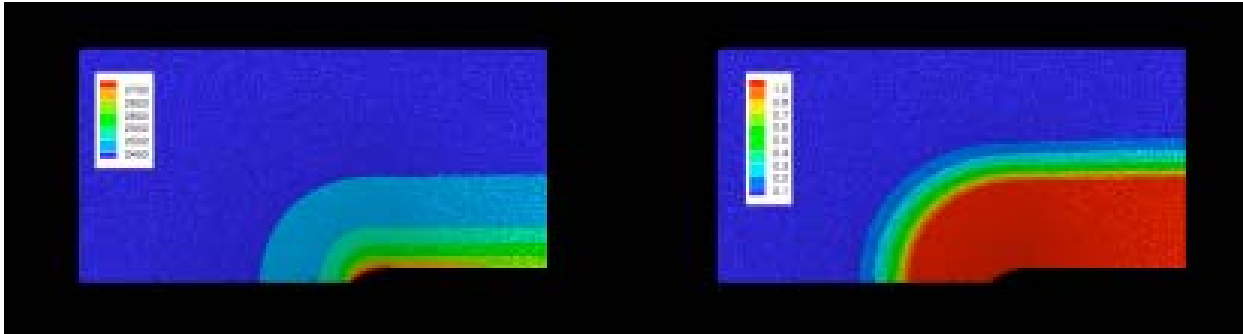


Figure 2. Left: the distribution of the density (kg/m^3) in the target at $t=0.925$ ms. Right: the distribution of damage in the target at $t=0.925$ ms.

Numerical simulations of penetration into concrete were performed. By conducting a time step sensitivity study it was shown that the numerical model is robust and computationally inexpensive. We discussed the evolution of the density changes around the penetration tunnel, of the shape and location of the rigid/plastic boundary, of the compaction zones and of the extent of damage due to air void collapse in such materials. For the constants involved in the model (shear and volumetric viscosities, cut-off yield limit and exponential weakening parameter for friction) that cannot be determined from data, a parametric study was performed.

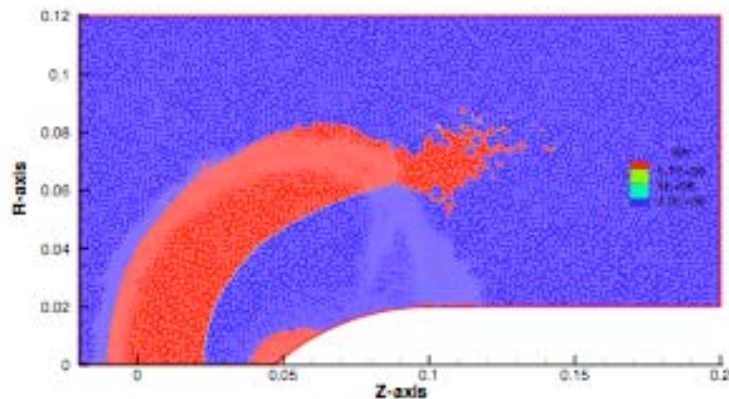


Figure 3. The distribution of the normal stress acting on the planes Ozr in the target. The regions where tensile failure occurs are colored in red. Since the tensile failure could occur only along planes which could not be symmetric with respect to the penetrator centerline, anisotropy is generated in the target. Trajectory deviations may be related to this induced anisotropy.

Keywords: decomposition-coordination formulation, augmented lagrangian method, high velocity penetration in porous materials, compressible rigid visco-plastic fluid, rate-effects, compaction, damage

- [1] O. Cazacu and I.R. Ionescu (2006), Compressible rigid visco-plastic fluids, *J. of Mech. Phys. of Solids*, Vol. 54, issue 8, 1640--1667.
- [2] O. Cazacu, I. R. Ionescu and T. Perrot, Steady-state flow of compressible rigid-viscoplastic media, *Int. Journal of Engng. Sciences*, vol 44 (2006), 1082--1097
- [3] O. Cazacu, I. R. Ionescu and T. Perrot, Numerical modeling of projectile penetration into compressible rigid visco-plastic media, *Int. J. Numer. Meth. Engng.* (2007), in press